CaO_2 depends on SaO_2 and hemoglobin, 1 g of hemoglobin carries 1.36 mL of O_2 (1.34mL - 1.39mL).
Oxygen content equals the oxygen combined (with hemoglobin), usually ~ 97% and the Oxygen dissolved (in plasma), usually ~ 3%.

Resistance = Pressure/Flow.
Pressure= Flow **x** Resistance.
Flow= Pressure/Resistance.

Cardiac

Cardiac	Abbrev	Formula	Normal value	Variables
Cardiac Index	CI	$CI = \frac{CO}{BSA}$	2.8-4.2L/min/m^2	CI = Cardiac Index (L/min/m2) CO = Cardiac Output (L/min) BSA = Body Surface Area (m2) **CI is the cardiac output indexed to BSA.
Cardiac output: Fick principle	CO	$CO = \frac{\dot{V}O_2}{C(a-\bar{v})O_2}$	4-5L/min	CO_fick = Cardiac Output (Fick) (L/min) VO2 = Oxygen Consumption (mL/min) C(a-v)O2 = a-v Oxygen Content Difference (B, mL/dL) *C(a-v)O2 is multiplied by 10 to adjust mL/dL to mL/L. **CO is the amount of blood pumped by the heart per minute.
Cardiac Output: Physiologic	CO	$CO = HR \times SV$	5-6L/min	CO_phys = Cardiac Output (Physiologic) (L/min) HR = Heart Rate (bpm) SV = Stroke Volume (mL/beat) *CO result in mL/min is divided by 1000 to adjust units to L/min. **CO is the amount of blood pumped by the heart per minute.
Coronary perfusion pressure	CPP	$CPP = DP - PCWP$	60-100 mmhg	CPP = Coronary Perfusion Pressure (mmHg) DBP = Diastolic Blood Pressure (mmHg) PCWP = Pulmonary Capillary Wedge Pressure (mmHg) **PCWP is also referred to as PAoP (Pulmonary artery occlusive pressure).
Ejection fraction	EF	$EF = \frac{SI}{EDV}$	60-70%	EF = Ejection Fraction (fraction) SI = Stroke Index (mL/beat/m2) EDV = End-Diastolic Volume (mL/m2)
Ejection fraction # 2	EF	EDV / ESV		
Maximal Allowable Blood loss	MABL	EBV x (Starting Hgb - Target Hgb / Starting Hgb)		EBV = Estimated blood volume Hgb = Hemoglobin *Some texts use Hct instead of hemoglobin
Mean arterial pressure	MAP	$MAP = \frac{SP + 2DP}{3}$	65-100	MAP = Mean Arterial Pressure (mmHg) SBP = Systolic Blood Pressure (mmHg) DBP = Diastolic Blood Pressure (mmHg) SP must be >= DP
Mean arterial pressure # 2	MAP	(CO x SVR) / 80 + CVP		CO = Cardiac output SVR = Systemic vascular resistance CVP = Central venous pressure
Mixed venous oxygen content	CvO2	CvO2 = (Hgb x 1.34) (SvO2/100) + (PaO2 x 0.003) x CO x 10	12-15ml/dl	CvO2 = mixed venous Oxygen Content (mL/dL) Hb = Hemoglobin (B, g/dL) SvO2 = mixed venous Oxygen Saturation (%) PvO2 = mixed venous Oxygen Tension (mmHg) *1g of Hb binds 1.36mL of O2.
Oxygen carrying capacity	CaO2	$CaO_2 = (Hb \times 1.36)\frac{(SaO_2)}{100} + (PaO_2 \times 0.0031)$	16-25mlO2/dL	CaO2 = arterial Oxygen Content (mL/dL) Hb = Hemoglobin (B, g/dL) SaO2 = arterial Oxygen Saturation (%) PaO2 = arterial Oxygen Tension (mmHg) *1g of Hb binds 1.36mL of O2.
Oxygen consumption	VO2	VO2 = CO x Ca - vO2 x 10	250ml/min	VO2 = Oxygen Consumption (mL/min) CO = Cardiac Output (L/min) C(a-v)O2 = a-v Oxygen Content Difference (B, mL/dL) *Multiplying Ca-vO2 (mL/dL) by 10 adjusts to mL/L.
Oxygen Delivery	DO2	$\dot{D}O_2 = CO \times CaO_2 \times 10$	800ml/min	DO2 = Oxygen Delivery (mL/min) CO = Cardiac Output (L/min) CaO2 = arterial Oxygen Content (mL/dL) *Multiplying CaO2 (mL/dL) by 10 adjusts to mL/L.
Oxygen extraction rate	O2ext	$O_{2_ext} = \frac{C(a-\bar{v})O_2}{CaO_2}$	4-5ml/dl	O2_ext = Oxygen Extraction Rate (fraction) C(a-v)O2 = a-v Oxygen Content Difference (B, mL/dL) CaO2 = arterial Oxygen Content (mL/dL)
Oxygen extraction Ratio	ER	VO2 / DO2	25%	VO2 = Oxygen Consumption (mL/min) DO2 = Oxygen delivery.
Peripheral vascular resistance index	PVRI	$PVRI = \frac{(MPAP - WP) \times 79.92}{CI}$	30-240dyn*s/cm5*m2	PVRI = Pulmonary Vascular Resistance Index (dyn*s/cm5*m2) MPAP = Mean Pulmonary Arterial Pressure (mmHg) PCWP = Pulmonary Capillary Wedge Pressure (mmHg) CI = Cardiac Index (L/min/m2) *79.92 is a conversion term to equalize units. *PCWP can't be > MPAP. In general, PVRI= MPAP/CI
Pulmonary vascular resistive	PVR	$PVR = \frac{(MPAP - WP) \times 79.92}{CO}$	20-120 dyns/cm5	PVR = Pulmonary Vascular Resistance (dyn*s/cm5) MPAP = Mean Pulmonary Arterial Pressure (mmHg) PCWP = Pulmonary Capillary Wedge Pressure (mmHg) CO = Cardiac Output (L/min) *79.92 is a conversion term to equalize units. *PCWP can't be > MPAP. In general, PVR= MPAP/CO
Pulse pressure	PP	$PP = SP - DP$	20-70 mmHg	PP = Pulse Pressure (mmHg) SBP = Systolic Blood Pressure (mmHg) DBP = Diastolic Blood Pressure (mmHg) Also: PP = (MAP - DP)/0.33
Starling Forces	Qcb	$Q_{CB} = K_{CB}[(P_C - P_B) - \sigma_{CB}(\pi_C - \pi_B)],$		Q_{CB} = Filtration across the blood-brain barrier $P_C - P_B$ = Hydrostatic pressure difference $\pi_C - \pi_B$ = Colloid osmotic pressure difference K_{CB} = Filtration Coefficient σ_{CB} = Reflection Coefficient
Stroke Volume	SV	$SV = \frac{CO}{HR}$	70-90ml/beat	SV = Stroke Volume (mL/beat) CO = Cardiac Output (mL/min) HR = Heart Rate (bpm) *The CO input is in mL/min (or L/min multiplied by 1000 to adjust units to mL/min). **SV is the amount of blood pumped by the heart with each beat; Also: SV= EDV – ESV.
Systemic vascular resistance	SVR	$SVR = \frac{(MAP - CVP) \times 79.92}{CO}$	770-1500 dyns/cm5	SVR = Systemic Vascular Resistance (dyn*s/cm5) MAP = Mean Arterial Pressure (mmHg) CVP = Central Venous Pressure (mmHg) CO = Cardiac Output (L/min) *79.92 is a conversion term to equalize units. *CVP must be < MAP. (may use Right Atrial Pressure in lieu of CVP). In general, SVR= MAP/CO
Systemic vascular resistance index	SVRI	$SVRI = \frac{(MAP - CVP) \times 79.92}{CI}$	1970-2390 dyn*s/cm5*m2	SVRI = Systemic Vascular Resistance Index (dyn*s/cm5*m2) MAP = Mean Arterial Pressure (mmHg) CVP = Central Venous Pressure (mmHg) CI = Cardiac Index (L/min/m2) *79.92 is a conversion term to equalize units. *CVP must be < MAP. (may use Right Atrial Pressure in lieu of CVP). In general, SVRI= MAP/CI

Increased gradient: diffusion defect, Right-To-Left shunt (intrapulmonary or intracardiac), V/Q mismatch.
Diffusion defect increases gradient on 15% O_2 but not on 100% O_2.
To rule out R to L shunt (right to left) , administer O_2 for 10 or 15 minutes, then recalculate A-aDO_2. In general, diffusion defects and mild to moderate V/Q defects correct (at least partially). Shunts do not correct.

Compliance = Volume/Pressure.

Respiratory

Respiratory	Abbrev	Formula	Normal value	Variables
A-a gradient Δ	A-a	$P_{A-aO_2} = P_{AO_2} - P_{aO_2}$ $P_{AO_2} = F_{iO_2} \times (P_B - P_{H_2O}) - \frac{P_{aCO_2}}{R}$ P_{aO_2} = measured	5-25mmHg	PA-aO2 = A-a Gradient (mmHg) FIO2 = Fraction of inspired O2 (fraction) PB = Barometric Pressure (mmHg) PH2O = Water vapor (partial pressure) (mmHg) R = Respiratory Quotient (fraction) PaCO2 = arterial CO2 Tension (mmHg) PaO2 = arterial Oxygen Tension (mmHg)
Airway resistance	Raw	$R_{AW} = P_{peak} - P_{plateau}$	1.5-2.0 cmH2O/L/S	Raw = Airway Resistance (cmH2O/L/s) Peak Pressure (cmH2O) Plateau Pressure (cmH2O) *Peak pressure must be > plateau pressure. Compliance = Volume/Pressure.
Alveolar gas equation	PAO2	$P_{AO_2} = F_{iO_2} \times (P_B - P_{H_2O}) - \frac{P_{aCO_2}}{R}$	95-105 mmHg	PAO2 = Alveolar Oxygen Tension (mmHg) FIO2 = Fraction of inspired O2 (fraction) PB = Barometric Pressure (mmHg) PH2O = Water vapor (partial pressure) (mmHg) R = Respiratory Quotient (fraction) PaCO2 = arterial CO2 Tension (mmHg)
Alveolar ventilation	VA	$\dot{V}_A = \frac{0.863 \times \dot{V}CO_2}{PaCO_2}$	3-5L/min	VA = Alveolar Ventilation (L/min) VCO2 = Carbon Dioxide Output (mL/min) PaCO2 = arterial CO2 Tension (mmHg) *VA is the total gas that reaches gas exchange units per minute. VA = VE - VD (all flows).
Dead space to tidal volume ratio (Bohr Equation)	Vd/Vt	$\frac{V_D}{V_T} = \frac{(PaCO_2 - P\bar{E}CO_2)}{PaCO_2}$	25-35%	VD:VT = Dead Space / Tidal Volume Ratio (%) PaCO2 = arterial CO2 Tension (mmHg) PECO2 = Expired CO2 Tension (mmHg) *PACO2 = PaCO2.
Dead space ventilation	VD	$\dot{V}_D = \frac{\dot{V}E \ (PaCO_2 - P\bar{E}CO_2)}{PaCO_2}$	1-3L/min	VD = Dead space Ventilation (L/min) VE = Minute Ventilation (L/min) PaCO2 = arterial CO2 Tension (mmHg) PECO2 = Expired CO2 Tension (mmHg) *VD is the total volume that ventilates the airways, non-perfused lung units and the ventilation tubing. VD = VE – VA (all flows).
Dynamic compliance	Cdyn	$C_{DYN} = \frac{V_T}{(peak\ pressure - PEEP)}$	60-100 ml/cmH2O	Cdyn = Dynamic Compliance (mL/cmH2O) VT = Tidal Volume (L) Peak Pressure (cmH2O) PEEP = Positive end-expiratory pressure (cmH2O) *PEEP must be < peak pressure. Compliance = Volume/Pressure.
Estimated arterial oxygen tension	PaO2_e	P_{aO_2}: Age correction = 100 – (0.33 × age)	80-100 mmHg	PaO2_e = arterial Oxygen Tension (estimated) (mmHg) Age = Patient Age (years)
FEV1 predicted	FEV1	FEV1 predicted = M = (Ht*4.30) – (Age*0.029) – 2.49 F = (Ht*3.95) – (Age*0.025) – 2.60	1.5-5.5 L	FEV1, predicted = Forced Expiratory Volume 1s (predicted) (L) Age = Patient Age (years) Sex = Gender (Male) Ht = Body Height (cm) The equations take Height in m (conversion is built in). Formulas by the Association for Respiratory Technology and Physiology (www.artp.org.uk).
FEV1/FVC	FEV1/FVC	$FEV_1/FVC = \frac{FEV_1}{FVC}$	70-120%	FEV1/FVC = Forced Expiratory Volume 1s/Forced Vital Capacity (%) FEV1 = Forced Expiratory Volume 1s (measured) (L) FVC = Forced Vital Capacity (L) Both are measured values.
Inspired Oxygen tension	PIO2	$P_{IO_2} = F_{IO_2} \times (P_B - P_{H_2O})$		PIO2 = Inspired Oxygen Tension (mmHg) FIO2 = Fraction of inspired O2 (fraction) PB = Barometric Pressure (mmHg) PH2O = Water vapor (partial pressure) (mmHg)
Minute ventilation	VE	$\dot{V}E = V_T \times RR$	4-8L/min	VE = Minute Ventilation (L/min) VT = Tidal Volume (L) RR = Respiratory Rate (resp/min) *VE is the total flow exhaled per minute. VE = VA + VD (all flows).
Physiologic dead space	VD	$V_D = \frac{V_T \ (PaCO_2 - P\bar{E}CO_2)}{PaCO_2}$	80-300mL	VD = Dead Space (physiologic) (mL) VT = Tidal Volume (mL) PaCO2 = arterial CO2 Tension (mmHg) PECO2 = Expired CO2 Tension (mmHg) *VD is ~ 1 mL per lb of Body Wt.
Physiologic Shunt	Qs/Qt	$\frac{\dot{Q}S}{\dot{Q}T} = \frac{CcO_2 - CaO_2}{CcO_2 - C\bar{v}O_2} \times 100$	0-7%	Qs/Qt = Shunt, physiologic (%) CaO2 = arterial Oxygen Content (mL/dL) CvO2 = mixed venous Oxygen Content (mL/dL) CcO2 = capillary Oxygen Content (mL/dL)
Static compliance	Cst	$C_{ST} = \frac{V_T}{(plateau\ pressure - PEEP)}$	70-10 mL/cmH2O	Cst = Static Compliance (mL/cmH2O) VT = Tidal Volume (mL) Plateau Pressure (cmH2O) PEEP = Positive end-expiratory pressure (cmH2O) *PEEP must be < plateau pressure. Compliance = Volume/Pressure.
Transpulmonary pressure	TPP	Alveolar pressure - pleural pressure	-8	
Transthoracic pressure	TTP	Pleural pressure - atmospheric pressure		
Apnea Hypopnea index	AHI	$\frac{Number\ of\ Episodes\ of\ Apnea\ and\ Hypopnea}{Hours\ of\ Sleep}$	<5	

Lung volumes

Total lung capacity	TLC	TLC = IRV + VT + ERV + RV	5-7L	TLC = Total Lung Capacity (L) IRV = Inspiratory Reserve Volume (L) VT = Tidal Volume (L) ERV = Expiratory Reserve Volume (L) RV = Residual Volume (L) *TLC is the volume of air in the lungs at maximal inspiration. (Also, TLC= VC+RV).
Vital capacity	VC	VC = IRV + VT + ERV	4-5L	VC = Vital Capacity (L) IRV = Inspiratory Reserve Volume (L) VT = Tidal Volume (L) ERV = Expiratory Reserve Volume (L) *VC is the maximum volume expelled by a forceful effort after maximal inspiration. (Also, VC= IC+ERV. VC= TLC-RV).
Residual volume	RV	RV = FRC – ERV	1-2L	RV = Residual Volume (L) FRC = Functional Residual Capacity (L) ERV = Expiratory Reserve Volume (L) *RV is the volume that remains in the lungs after maximal expiration. (Also, RV= TLC-VC).
Inspiratory capacity	IC	IC = IRV + VT	1-4L	IC = Inspiratory Capacity (L) IRV = Inspiratory Reserve Volume (L) VT = Tidal Volume (L) *IC is the maximum volume inspired from resting expiration.
Functional residual capacity	FRC	FRC = ERV + RV	2-4L	FRC = Functional Residual Capacity (L) ERV = Expiratory Reserve Volume (L) RV = Residual Volume (L) *FRC is the volume remaining in the lungs after tidal volume (VT) is expired.
Total volume	VT	VT = Wt + VT	400-600mL	VT = Tidal Volume (mL) Wt = Body Weight (kg) VT = Tidal Volume (mL/kg) *VT is the volume of air inspired/expired with each breath. (Also, VT= IC-IRV. VT= VC-(IRV+ERV), VT= TLC-(IRV+ERV+RV).
Closing Capacity	CC	CC = CV + RV	1.3-2.5L	CC = Closing Capacity (L) CV = Closing Volume (L) RV = Residual Volume (L) *CC is the addition of closing volume + residual volume.

Physics laws

	Formula	Variables	Application
Boyles law	$P_1 \bullet V_1 = P_2 \bullet V_2$	P = pressure V = Volume	What gas law applied to know the volume of O2 in a full "E" type of cylinder available for use at 15 psi? At constant temp, volume of a given mass of gas is inversely proportional to the absolute pressure. We know the volume of an Etype of cylinder is approximately 5 L. The service pressure at which the cylinder is filled is 2000 psi 2000 x 5 / 15 = 665L
Charles law	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	V = Volume T = Temperature	Respiratory gas measurements of tidal volume & vital capacity etc are done at ambient temperature while these exchanges actually take place in the body at 37C One way of heat loss from the body is that air next to the body surface gets warmer and moves up and thus our patient loses heat this way (esp. important in paediatric anaesthesia).
Gay Lussac	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$	P = pressure T = Temperature	At constant volume, the absolute pressure of the given mass of gas is directly proportional to the temperature. APPLICATION: i. Medical gases are stored in cylinders having a constant volume and high pressures (138 Barr in a full oxygen / air cylinder). If these are stored at high temperatures, pressures will rise causing explosions. ii. Molybdenum steel can withstand pressures till 210 bars. Weakening of metal in damaged cylinders are at a greater risk of explosion due to rise in temperature.

Ideal gas law	Pressure • Volume = # moles • 0.0821 • Temperature
	PV= nRT
		P = pressure
		V = Volume
		T = Temperature
			Equal volume of gases contain equal number of molecules at standard temperature and pressure (273K and 760mm Hg). The law can also be defined as one gram molecular weight (one mole) of a gas contains 6.023x1023 (avogadro's number) molecules = occupies 22.4L at STP. PV = n RT – is the ideal gas equation. R is the universal gas constant = 1.987 J/degree/mole in SI units. APPLICATION: Since the cylinder volume is constant, temperature is constant and R is already a constant P = n, i.e. pressure shown in the Bourdon's gauge is proportional to the number of molecules which is the amount of gas in the cylinder. Hence the pressure gauge acts as a content gauge.

Fahrenheit to Celsius	$(F - 32) \cdot \frac{5}{9}$	F = Fahrenheit	Converting F -> C
Daltons Law	$P_{total} = P_1 + P_2 + P_3 \ldots$	P = pressure	In a mixture of gases, the total pressure exerted by the mixture is equal to the sum of the partial pressures of the individual gases, provided the gases donot mix with each other. P = p1+p2+p3….
Henderson - Hasselbalch	$pH = pK_a + \log \frac{[A^-]}{[HA]}$ $pH = pK_a + \log \frac{[B]}{[BH^+]}$	A = weak acid B = weak base	Degree of ionization
Henry's Law	Henry's Law: Partial Pressure = Constant • Concentration Henry's Law: [CO2]=pCO2 x solubility of CO2	Obv.	is a gas law that states that the amount of dissolved gas in a liquid is proportional to its partial pressure above the liquid.
Poiseuille Law	$\mathcal{F} = \frac{P_1 - P_2}{R}$ $R = \frac{8\eta l}{\pi r^4}$	F = flow rate P = pressure R = radius n = viscosity l = length of tubing	If you want to give blood rapidly… What will you do? 1.Put a wider gauge canula 2.Increase the drip stand height
Law of Laplace (cylinder)	Pressure x Radius / Wall thickness		Excess pressure inside a spherical gas-liquid interface is equal to twice the coefficient of surface tension divided by the radius of the interface.
Law of Laplace (Sphere)	Pressure x Radius / 2 x Wall thickness		
Reynolds Number	$Re = \frac{\rho v d}{\eta}$	Re – Reynold's number ρ – density of fluid v – velocity of fluid d – diameter of tube and η – viscosity of fluid	Reynold's number > 2000 – indicates turbulent flow <2000 – indicates laminar flow Why would you not use connectors with sharp curves? At the sharp bends the flow converts into a turbulent flow as the REYNOLD NUMBER will be more than 2000. This will increase the resistance to the flow. Every piece of anaesthetic equipment, because of diameters & shape of connectors, number & arrangement will effect FGF. Wide bore & curved rather than sharp angles should be preferred
Grahams law of diffusion	$\frac{R_1}{R_2} = \sqrt{\frac{M_2}{M_1}}$	R = Rate M = density	States that flow rate is Directly proportional to the square root of the pressure gradient on either sides of the tube Inversely proportional to the square root of the density of the fluid. APPLICATION If the anaesthesia machine is used in a high altitude area, where the atmospheric pressure is very low, the density of the gas decreases, but viscosity will not change. As higher flows depend on density and as per GRAHAM'S LAW FOR TURBULENT FLOW, flow is inversely proportional to square root of density i.e. Flow will be higher than the actual flows that are set in the flow meters. The opposite will occur under hyperbaric conditions.
Ohms law	*Current* = *Voltage Difference* / *Resistance* *Flow* = *Pressure Gradient* / *Resistance*		
Nernst Equation	E Ion = – 61.5 log ([Ion] inside / [Ion] outside)		Used to predict an ions equilibrium potential. Equilibrium is achieved when there is no concentration gradient and therefore no net flow of ion across the membrane
Ficks Law of diffusion	$\dot{V}_{gas} = \frac{D * (P_1 - P_2) * A}{T}$	Diffusion Coefficient (D) Partial pressure gradient ($P_1 - P_2$) Surface area (A) Thickness of barrier (T)	Diffusion of drugs to fetus

	Hemodynamic Term	Symbol
Flow	Cardiac Output	Q
Pressure Gradient	MAP – CVP	$P_1 - P_2$ or ΔP
Resistance	Systemic Vascular Resistance	R

System specific/Miscellaneous formulas

ETT size (uncuffed)	Uncuffed ETT Size = $\frac{Age}{4} + 4$
ETT size (cuffed)	Cuffed ETT Size = $\frac{Age}{4} + 3.5$
Fraction of inspired oxygen	$FiO_2 = \frac{(Oxygen\ Flow\ Rate \bullet 100) + (Air\ Flow\ Rate \bullet 21)}{Total\ Flow\ Rate}$
Cerebral perfusion pressure	Cerebral Perfusion Pressure = MAP – ICP (or CVP whichever is higher)
Anion Gap	Anion Gap = $[Na^+] - ([Cl^-] + [HCO_3^-])$
Plasma osmolarity	Plasma Osmolarity = $2[Na^+] + \frac{Glucose}{18} + \frac{BUN}{2.8}$
BMI	$\frac{Weight\ (kg)}{Height\ (m^2)}$
IBW	*Men (kg) = Height (cm) - 100* *Women (kg) = Height (cm) - 105*
LBW	*IBW x 1.3*
Volume of distribution	Volume of Distribution = $\frac{Dose}{Concentration}$
Half life	Half-life = $\frac{Volume\ of\ Distribution}{Rate\ of\ Clearance}$
Loading dose	$Vd \times \frac{Desired\ Plasma\ Concentration}{Bioavailability}$
Rule of nines	• Head = 10 percent • Trunk = 36 percent • Arm = 9 percent • Leg = 18 percent • Perineum = 1 percent
Modified Brooke formula	*2 mL x % TBSA burned x kg (body weight)*
Parkland formula	*4 mL x % TBSA burned x kg (body weight)*
Uterine blood flow	UBF = uterine arterial pressure – uterine venous pressure / uterine vascular resistance
Hypotension in neonate	[(70) + (age in years x 2) mmHg
GFR	*GFR = [(140 - age) x Body Weight (kg)] / [72 x Serum Cr (mg/dL)]*

Source:
MediCalc :: EQUATIONS :: by ScyMed :: http://www.scymed.com/en/smnxfd/smnxfdah.htm.
Accessed May 20, 2021.

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